

58 A student investigated the tadpole population in a large pond.

A net with a rectangular opening measuring $0.1\text{ m} \times 0.2\text{ m}$ was swept through the water for a fixed distance of 1 m. This was repeated 10 times.

All the sweeps were made at the edge of the pond as the student had no waders or boat.

The number of tadpoles in each sweep was recorded in the table.

<i>sweep number</i>	1	2	3	4	5	6	7	8	9	10
<i>number of tadpoles</i>	20	12	32	0	4	8	4	8	12	20

The student made the following statements.

- 1** Each sweep sampled 0.02 m^3 of water.
- 2** The frequency of occurrence of the tadpoles was 90%.
- 3** An accurate estimate of the population size of tadpoles in the pond could be calculated using this data if the total volume of water was known.

Which of the statements about the investigation is/are correct?

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3